

Automatic Privacy Policy Compliance Assessment Leveraging NLP Models

1st Cătălina-Mihaela Priescu
The Faculty of Automatic Control
and Computers
Politehnich University of
Bucharest
Bucharest, Romania
catalina.priescu@stud.acs.upb.ro

2nd Mihnea-Alexandru Moisescu
The Faculty of Automatic Control
and Computers
Politehnich University of
Bucharest
Bucharest, Romania
mihnea.moisescu@upb.ro

3rd Miruna-Elena Iliuță
The Faculty of Automatic Control
and Computers
Politehnich University of
Bucharest
Bucharest, Romania
miruna_elena-iliuta@upb.ro

Abstract - Following its adoption in 2018, the General Data Protection Regulation (GDPR) has impacted data processing entities significantly, especially by compelling them to ensure their privacy policies are compliant with a defined set of requirements. Manual assessment of GDPR compliance is laborious and susceptible to errors, driving increased research interest in automated solutions with such capabilities. This paper provides a systematic review of the recent advancements in the automated evaluation of privacy policies employing Natural Language Processing (NLP) methods. Within this context, this study proposes a structured approach for processing Romanian privacy policies, a particularly complex task, considering both the low-resource status of the Romanian language and the intricacy of the regulatory wording utilized. Aiming to address these constraints, this paper proposes a compliance verification system, leveraging the semantic similarity capabilities of the E5 Transformer model. Experimental results indicated that the trained model achieved a 75% accuracy in matching sentences extracted from a Romanian privacy policy to the appropriate GDPR requirements. Furthermore, the paper tackles the challenges of adapting NLP-based solutions to other languages with scarce resources, providing insights and directions for future research in multilingual and domain-specific settings.

Keywords: GDPR Compliance, Privacy Policies, NLP, Semantic Similarity Analysis, Low-resource Languages, E5 Transformer Model.

I. INTRODUCTION

As digital services continue to expand, the amount of personal information transmitted online has grown substantially. This has led to increased interest in how to properly process personal data while protecting users' privacy. Consequently, data protection regulations were established, which granted users rights over their personal data and provided organizations guidelines for legitimate data processing techniques. In 2018, the European Union adhered to this trend and adopted the General Data Protection Regulation, which is considered one of the most rigorous data processing frameworks [20].

While often perceived as an administrative formality, a privacy policy is a document enforced by data processing regulations. To that end, the GDPR requires organizations to clearly describe their data processing activities as well as outline the rights users hold over their personal data within the privacy policies. However, ensuring that a privacy policy is addressing all the intricacies posed by these regulatory requirements can be challenging. Especially in the Romanian context, both the interpretation and the implementation of

GDPR provisions are particularly difficult due to the shortage of domain experts within this field [22].

At policy level, nonconformities can arise from several factors that include, but are not be limited to, human errors, insufficient understanding or misinterpretation of GDPR requirements. Literature indicates that noncompliance is primarily caused by failing to include several data processing activities within the policy, third party data sharing being the most common [1]. Additionally, this paper revealed that while Romanian organizations acknowledge users' rights within their policies, they rarely explain how these rights can be exercised or what restrictions may occur in doing so, which contradicts GDPR's principles. These compliance issues usually persist in time, as the standard does not require a periodic review of the privacy policy. Moreover, when such revisions take place, they are usually carried out by the same person who issued the policy [2]. In such case, this practice may result in a biased and unreliable assessment of the compliance level.

Recognizing these challenges, this study investigates whether semantic similarity capabilities of NLP models can be leveraged to support the automated assessment on GDPR compliance of privacy policies written in Romanian. Unlike classic methods that rely on keyword identification, semantic models can detect references to GDPR requirements, even when paraphrased. This research focuses on addressing two main challenges posed by this compliance task: the low-resource status of Romanian and the complexity of legal terminology used in privacy policies. In doing so, this paper contributes to the current developments in employing natural language processing models for automated compliance assessment in low-resource settings.

At a functional level, we proposed a system that fully supports the automated compliance evaluation. This is achieved by employing the semantic similarity capabilities of NLP models to compare each sentence extracted from a privacy policy to each GDPR requirement found relevant. Following this analysis, the system automatically generates a compliance report that includes an overall compliance score derived from the number of GDPR requirements addressed by the policy, along with a set of recommendations on how to address the ones that are missing. Furthermore, the system generates a metric that quantifies the level of ambiguity of the policy text that acts as an indicator of excessive use of vague

terms. This can lead to a decreased assessment accuracy and may further indicate an instance of GDPR noncompliance, considering that the framework mandates the use of accessible language in presenting data processing practices within privacy policies [19].

The remaining sections are organized as follows. Section II outlines the recent advancements in the automation of compliance assessments leveraging NLP technologies. Section III describes the process followed in designing and implementing the automated tool. Section IV summarizes the experimental results derived from a case study used to validate the proposed system. Section V provides an overview of the key contributions of this study, critically examines the limitations. Finally, section VI presents the conclusions and proposes directions for future research.

II. CURRENT ADVANCEMENTS IN COMPLIANCE ASSESSMENT AUTOMATION

This section provides a systematic review of the scientific studies that investigate the application of NLP technologies in automating compliance assessments with data protection regulations. This analysis further identifies the functionalities proposed by researchers to enhance automated compliance evaluation such as compliance scoring or automated reporting, as well as the appropriate training methodologies for semantic models. Following keyword-based screening (e.g. “NLP”, “GDPR compliance”, “Privacy Policy”, “Semantic Similarity Models”, “Automatic Compliance Assessment”) and examining the abstracts, we selected several studies that we considered relevant to this research, and we classified them as follows.

A. Early Approaches: Rule-Based Methods

In order to assess the level of compliance of a policy, Asif et al. introduced a system in [3] that automatically gathers privacy policies collected from various local websites and tracks the occurrence of predefined phrases which refer to data processing principles throughout the text. In line with this methodology, Kim et al. [5] utilized a rule-based text analysis system that identifies expressions which denote data processing practices and aggregates them to assign a compliance score to the policy. In a further step, Mousavi et al. [39] introduced a functionality that automatically generates a compliance report for each policy, that contains the number of expressions referencing data handling principles found within the policy.

Sánchez et al. extended this approach [4] by replacing the identification of solely keywords with extended phrases, by using bag-of-words methods. In this research, each privacy policy was automatically analyzed using a support vector machine trained to annotate said phrases with labels denoting data processing techniques, based on which a compliance score was generated. A significant contribution is brought by Aberkane et al. in [36], who extended the automated compliance analysis by applying it to a large corpus of privacy policies. Researchers used classifiers to automatically assess the compliance level of over 8000 policies, based on the identification of five key terms related to data processing principles across the documents. The findings were corroborated with internal company

information to determine the potential causes of the identified nonconformities.

B. Modern Developments: Transformer-Based Regulatory Analysis

Del Alamo et al. [6] conducted a literature review and identified a trend in adopting Transformer models to perform NLP tasks. They recognized BERT as the most widely used model for automated compliance analysis applications. BERT models hold the capability to evaluate sentences bidirectionally, thereby analysing each word based on the context in which it appears. Extending this model, Zhang et al. [7] trained a BERT model to perform as a multi-label classifier, using predefined labels, to identify GDPR requirements within privacy policies.

In [8], Liu et al. proposed merging BERT technologies with automatic entity recognition methods to ascertain relevant GDPR passages in policies. This hybrid approach demonstrated an improvement in the detection accuracy of segments that reference data processing methods. Extending this approach, Manandhar et al. [9] presented a model that extracts the data processing requirements that are recurring across different standards (i.e. not limiting the study to GDPR) and checks whether they are found at the privacy policy level.

C. Emerging Techniques: Semantic Similarity Analysis

Elluri et al. [10] explored the use of cosine distance to measure the semantic similarity between sentences extracted from privacy policies and GDPR requirements. Based on this similarity value, the researchers yielded a compliance score for each policy. To adapt this approach to a more precise analysis, Amaral et al. [11] proposed a system that aggregates each cosine similarity score between extracted sentences and GDPR requirements in order to estimate to what extent the policy addresses the requirements stipulated in the regulation.

To further optimize the process, Torre et al. trained traditional classifiers to automatically label sentences from the privacy policy according to GDPR provisions. The innovation in their study [26] lies within the choice to switch from standard sentence embeddings to GloVe vectors for calculating the cosine similarity. The applicability of semantic similarity is further extended to studies aiming for dynamic compliance analysis as depicted in [13]. McConkey and Olukoya leveraged semantic similarity to measure inconsistencies between statements extracted from privacy policies and the permissions required at the code level by Android mobile applications.

Lastly, a significant advancement in this field arises from the applications of these technologies in the multilingual setting [12]. Wang et al. [14] enlarged Transformer’s capabilities by developing Multilingual-E5, which enables semantic similarity assessment across over 100 languages, including Romanian. Niculae et al. demonstrated the model’s capabilities in Romanian by applying it in the medical domain [15]. This suggests that such methods may also be effective in the legal context, where terminology is generally less complex.

D. Research Gaps

Following the review of the relevant literature, a significant gap was identified in the automation of GDPR compliance, specifically the assessment of privacy policies written in Romanian. Although existing studies propose various methodologies based on NLP, even in multilingual settings, the literature revealed no research on applying these approaches to policies in Romanian. Hence, the aim of this paper is to investigate the use of semantic similarity models in automated GDPR compliance evaluation of Romanian privacy policies.

Furthermore, most of the studies that were analyzed, relied on the OPP-115 corpus [21], a collection of annotated privacy policies with data processing information by legal experts, for training and assessment. By using this data set, researchers have been able to improve the performance of models for compliance automation tasks in regulatory settings. Given that no similar annotated dataset exists for Romanian, training effectively the NLP model in this context remains challenging. To address this resource limitation, a dataset consisting of Romanian privacy policies annotated according to GDPR provisions was developed for this research.

Hence, the research direction proposed by this study is the investigation of how the semantic similarity capabilities could be leveraged to successfully identify the GDPR compliance level of privacy policies. Essentially, we propose testing whether these models can overcome the constraints imposed by legal terminology and limited resources.

III. DESIGN AND IMPLEMENTATION OF A NLP-BASED COMPLIANCE ASSESSMENT MODEL

A. Structuring Legal Requirements

To support the automated GDPR compliance assessment of privacy policies, we developed a customized knowledge base. This was achieved by analyzing the GDPR standard and extracting only those requirements that address the privacy policy contents. Each article found relevant (which will be referred to as a requirement in this study) was classified as either:

- Mandatory: address all privacy policies;
- Conditional: applicable only under specific organizational conditions (e.g. NGO status of an institution).

Subsequently, each requirement is recorded within the knowledge base, in JSON format, along with four corresponding attributes: a unique identifier *rule_id*, *mandatory* which indicates its applicability status, a *description* field that contains the regulatory provision adjusted to reflect typical Romanian privacy policy phrasing, and finally a *recommendation* field that contains guidance on how to properly address the requirement, if identified as missing. Hence, each requirement is represented as follows:

```
"rule": [
  {
    "rule_id": "",
    "mandatory": "",
    "description": "",
    "recommendation": ""
  }
]
```

To ensure an accurate assessment process within the system, the compliance analysis begins with an eligibility questionnaire that dynamically selects the applicable GDPR requirements based on organizational context. The questionnaire consists of nine targeted questions, each corresponding to one of the GDPR articles classified as “conditional”. The responses determine which of the conditional requirements are applicable, thus creating a customized GDPR assessment, tailored to each organization. By adopting this approach, we prevented the system from evaluating policies against inappropriate requirements.

B. Model Selection Strategy

An exploratory assessment was conducted to determine whether existing pre-trained Transformer models could effectively support the specific criteria of this endeavor. The analysis focused on models that had the following features: Romanian language pre-training and semantic similarity assessment capabilities. After reviewing the existing state-of-the-art solutions, two models revealed these characteristics: *iliemihai/sentence-bert-base-romanian*, *intfloat/multilingual-e5-base*. To assess between them, we chose to evaluate their baseline performance in the compliance context. Hence, the models were tested on a similarity assessment task.

The dataset used for this evaluation consisted in pairs of sentences representing GDPR user rights: access, deletion, objection and portability. Each set contained one sentence semantically similar to the user’s right, one contradictory (i.e. denied the right), and one neutral (i.e. contextually irrelevant to the right), designed to assess the models’ ability to discriminate between semantically similar and contradictory statements that often use the same words.

Results revealed (Fig.1) that while *sentence-bert-base-romanian* consistently overestimated semantic similarity across all text pairs, *multilingual-e5-base* showed modest ability to distinguish between semantic and contradictory statements. This is reflected in the slightly smaller similarity scores for the contradictory pairs, making it more suitable for compliance classification tasks, since they require sensitivity to semantic variations. Hence, the latter was selected for use in the subsequent stages of this research.

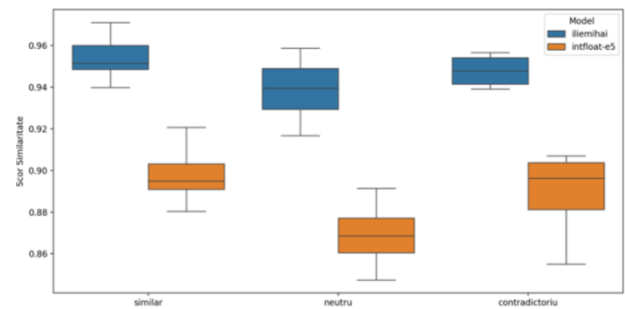


Fig.1. The zero-shot analysis results of the candidate models

C. Unsupervised Training

Literature supports the pre-training of Transformer models on domain-specific texts, as it has been demonstrated to improve

the model's ability to recognize specialized terminology after sufficient exposure to relevant texts [16]. Building on these findings, an intermediate unsupervised training task was carried out to adapt the Multilingual-E5 Transformer model to the privacy policy terminology.

The pre-training corpus consisted of over 20,000 sentences extracted from Romanian privacy policies across diverse sectors (financial, healthcare, education, and e-commerce), ensuring extensive lexical diversity. This corpus size is supported empirically in [17], where it is suggested that a moderately sized domain-specific corpus can lead to a noticeable increase in performance. During this pre-training phase, the model improved steadily, reducing loss values consistently from an initial value of 9.54 to 2.03 in the final epochs. This descending trend confirms a favorable execution of the unsupervised learning phase [18].

D. Supervised Training

It was previously emphasized that carrying out a fine-tuning process is essential to enhance the model's capacity to differentiate between semantically similar and contradictory sentences. Refining this ability helps to reduce the risk of inaccurate matches caused by accidental lexical overlap. To support this training approach, a data set containing similar sentence pairs was manually established. Each pair consisted of a GDPR article description from the knowledge base and a sentence extracted from Romanian privacy policies that addressed the regulatory requirement. To ensure compatibility with the architecture of the *intfloat/multilingual-e5-base* model, the data used for the fine-tuning process was structured following a "query" and "passage" format. In this context, *query* corresponds to the GDPR article, while *passage* refers to the sentence used as an example.

Upon a review of training methodologies presented in the literature, Cosine Similarity Loss (CSL), Triplet Loss (TL), and Multiple Negatives Ranking Loss (MNLR) were identified as the most suitable strategies in fine-tuning Transformer models performing semantic similarity tasks. To empirically establish the most effective approach in this specific context, a comparative analysis was performed using the base E5 model and its three variants trained with each of the methodologies that were previously mentioned. The evaluation was conducted on an extended dataset of Romanian sentence pairs containing either similar or contradictory statements of GDPR user rights and data processing principles, to assess each model's ability to discriminate in realistic scenarios.

As shown in Figure 2, the model fine-tuned using CSL (orange) demonstrated the most substantial improvement, producing a clear separation between similar pairs (median similarity score of 0.8) and contradictory pairs (median of 0.5, skewed toward lower values), indicating enhanced semantic sensitivity. In comparison, the MNLR model (green) achieved only moderate separation, while the TL-trained model (pink) failed to distinguish between the two classes, even overestimating similarity more than the untrained base model (blue). These results highlight CSL as the most effective training strategy in this context. However,

limitations such as insufficient training data, especially for contradictory examples and computational constraints limited the current results.

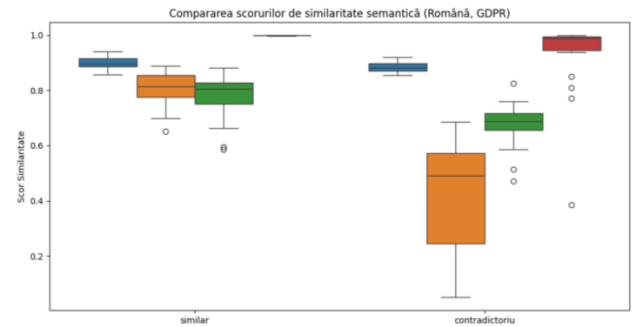


Fig.2. Results of CSL, TL and MNLR training methodologies against base model

E. System Integration

To demonstrate the practical applicability of the proposed methodology, the fine-tuned multilingual-E5 model was integrated into a dedicated web-based application designed to assess Romanian privacy policies for GDPR compliance. The system is implemented as a web application via Flask-based backend, Firebase cloud databases, and the semantic similarity model based on CSL fine-tuned multilingual-E5. Through this platform, users can upload a privacy policy and fill in the eligibility questionnaire that initiates the compliance analysis. Subsequently, the text extracted from the policy is automatically segmented into individual sentences using the *ro_core_news_sm* model from the *SpaCy* library, which is adapted to Romanian syntax. Each sentence is then compared to the requirements in the knowledge base that were found relevant using cosine similarity. If the sentence with the highest similarity score exceeds a predefined threshold, the corresponding requirement is considered addressed. The results of this assessment are outlined in an automatically generated compliance report, which includes an overall conformity score reflecting the extent to which the policy addresses applicable GDPR requirements. This score is represented by the ratio between the number of GDPR requirements identified within the policy and the number of requirements deemed relevant considering the organizational context.

In addition to the conformity score, the system calculates an ambiguity metric that quantifies the occurrences of vague Romanian terms found within text. This component was introduced to address Article 12 in GDPR, which mandates that policies must be written in "clear, concise, and understandable language" [19]. The ambiguity metric is computed using a predefined dictionary of vague expressions often encountered in Romanian privacy policies (e.g., "in certain cases", "where applicable", "we may use", etc.). To avoid counting the legitimate cases, an additional dictionary was defined containing exceptions, i.e. phrases that clarify the use of the vague expressions. Hence, if a sentence contains a vague term without an appropriate clarification, it counts towards the overall ambiguity score.

Additionally, when different versions of a privacy policy are uploaded, the platform automatically saves both conformity and ambiguity scores and tracks their evolution over time. These results are displayed as visual graphs that can be downloaded alongside with the compliance report. This feature enables iterative policy refinement, allowing users to implement recommendations, upload revised documents, and assess the changes in both compliance scores and textual clarity.

IV. EXPERIMENTAL RESULTS

The model's accuracy was assessed by conducting a case study involving a real Romanian privacy policy, selected at random. Each sentence in the policy was tested against all GDPR requirements in the knowledge base, regardless of their mandatory or optional status, to evaluate the model's semantic matching performance and analyze the matching patterns to gain insights into possible biases or errors.

In its default configuration, the system compares each sentence from the privacy policy against all applicable GDPR rules and assigns a match when for the highest cosine similarity value that exceeds a predefined threshold of 0.75. Once a sentence is matched to a rule, it is excluded from further comparisons to optimize the evaluation process. However, to assess the model's actual capability to compute semantic similarity, this feature was intentionally disabled for the course of this case study. Specifically, the similarity threshold was raised to 0.99 to prevent any matches from occurring, ensuring that all sentence requirements pairs were verified. Hence, we forced the system to compute similarity scores between each sentence and every individual rule. From these results, the highest similarity score per sentence was considered a semantic match and was compared against the expected annotation. The aim was to determine whether the rule associated with the highest score match was indeed the anticipated one, thus assessing the model's semantic matching capabilities in a fully unrestricted setting.

Under these controlled conditions, the model successfully extracted all sentences that reference data processing principles. However, when assessing the matching performance, the results indicate the partial success of this trained model, which correctly mapped 12 sentences out of 16, demonstrating a 75% success rate. Nevertheless, even for the improperly mapped instances, the model generated similarity scores very close to the match (differences of less than 0.1), indicating potential for improvement through more extensive training.

Moreover, when analysing of the four misclassified cases, it was revealed that these statements were longer and more ambiguous, whereas correctly identified ones were rather straightforward in their phrasing. This pattern reinforces that, with an expansion of the datasets, the model's performance could be significantly improved. Overall, the system successfully identified only the relevant sentences from the privacy policy and demonstrated functional capability across all features, validating the approach in practical contexts of GDPR compliance analysis.

V. DISCUSSIONS

This research offers several contributions to the field of compliance automation. Firstly, we conducted a literature review on existing approaches to automated compliance assessment, providing current insights into this domain. Secondly, this study established a fine-grained GDPR knowledge base that includes exclusively the requirements referencing privacy policies. To refine it at the implementation level, we developed a questionnaire to filter the applicable requirements according to organizational characteristics. Moreover, a corpus of annotated sentence pairs was created, consisting of real examples extracted from privacy policies mapped to relevant GDPR requirements. This resource further enabled model training and performance evaluation.

Additionally, multiple Transformer models were tested in zero-shot conditions, to determine which performed best for Romanian-language semantic similarity tasks. Based on the comparative performance analysis, the Multilingual E5 model was selected as it showed potential for further development. Subsequently, fine-tuning methodologies were researched and three of them were systematically applied to determine the most effective training approach for adapting the selected model to GDPR compliance assessment.

Finally, the research resulted in the development of a fully functional application that supports key functionalities for assessing GDPR compliance. Specifically, these include estimating the level of textual ambiguity, assessing the compliance level of the policy and identifying the requirements that were not found in the policy. For each missing requirement the system provides specific recommendations on how to be addressed to prevent future nonconformities. All these details are included in a compliance report that is automatically generated. Moreover, the system tracks the scores' evolution over time, which enables analysis between policy versions. These contributions address both technical and practical aspects of processing natural language of a regulatory domain in a low-resource language. Therefore, this study demonstrated the applicability of semantic similarity methods to the Romanian language.

Despite encouraging results, this research was subject to several limitations. Firstly, the system was created based on the authors' heuristics, which pose several risks such as subjectivity, bias, and the potential of it being built on a deficient foundation, due to inaccurate understanding of the GDPR regulatory requirements. Secondly, the reduced size of the annotated corpus also represented a substantial limitation, as Transformer models typically require an enlarged training dataset to achieve robust performance. In addition to that, the annotation task should be carried out manually, hence it was subject to human error. Moreover, the dataset lacked adequate contradictory examples, since such cases rarely occur in real privacy policies.

Another significant limitation arises from the complexity of language used both in the privacy policies and in the GDPR provisions. Hence, we consider the legal terminology a

limitation as it is challenging to capture all its intricacies through semantic similarity alone. Collectively, these limitations underline the dependency of the model's performance on multiple factors and suggest that additional research should be carried out focusing on expanding the dataset, incorporating domain-expert annotation, and as previously stated, retraining the model under improved technical conditions.

VI. CONCLUSIONS

This study explored the potential of natural language processing capabilities to automate the assessment of privacy policy compliance with the General Data Protection Regulation in Romanian, which is considered a low-resource language. By addressing such outstanding gap in the existing literature, this research proposed and empirically validated a methodology to evaluate semantic similarity between statements extracted from a privacy policy and GDPR requirements in order to assess its level of compliance. Based on the Multilingual E5 Transformer, the CSL fine-tuned version of the model was integrated into a web-based tool that allows users to upload privacy policies and automatically issue a compliance report on it. The results demonstrate the efficiency of using semantic similarity for automated compliance analysis and highlight the possibility of leveraging this approach in limited-resourced languages. Furthermore, the findings indicate that the model's performance could be enhanced through proper training.

In addition, several further enhancements are planned to improve the functionality of the proposed system. One significant improvement is implementing an automatic contradiction detection mechanism that could be used to identify inconsistencies within the text. Similarly, such advanced language models could be leveraged to automatically identify textual ambiguities, thereby enhancing the static assessment method proposed by this research. Moreover, we propose the implementation of a system that automatically updates the GDPR knowledge base to reflect any changes to the regulatory framework. This approach could be accomplished by employing a crawling and parsing mechanism. Specifically, the crawling component would monitor the official statements issued by regulatory bodies and, subsequently, the parsing mechanism would automatically incorporate them into the knowledge base. This approach would ensure continual alignment with evolving compliance requirements while minimizing manual maintenance work.

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